

AI DOCUMENT FORENSIC VERIFICATION CERTIFICATE

Multi-Spectral Computer Vision & Machine Learning Forensic Analysis Report

Document UUID: DOC-FFD947A348	Verification Verdict: FORGED
File Name: sample_certificate.jpg	Forensic Confidence: 99.0%
Analysis Date: 2026-08-21 04:11:13 UTC	Risk Level: HIGH
Inference Pipeline: MultiSpectral-Forensic-v2.0 (8-Module Ensemble // Profile: FULL)	Primary Tampering Category: Generative AI & Inpainting Synthesis
Processing Latency: 0.27s	Suspicious Anomaly Zones: 0

DETECTED FORGERY CATEGORIES BREAKDOWN

Category Taxonomy	Severity	Confidence	Forensic Explanation
Generative AI & Inpainting Synthesis	HIGH	99.0%	2D Fast Fourier Transform (FFT) reveals synthetic spectral grid harmonics typical of Generative Diffusion models.

8-MODULE SPECIALIZED FORENSIC AUDIT SCORECARD

Analysis Module	Status	Risk Level	Score	Key Findings Summary
Visual & Compression Discrepancy Module	PASS	LOW	89.7%	Uniform JPEG compression grid consistent with original camera/scan capture.
Noise Variance & Sensor Fingerprint Module	TAMPERED	HIGH	98.5%	Elevated spatial noise inconsistency score (2.050 vs normal <0.85)
Copy-Move & Spatial Cloning Module	PASS	LOW	95.0%	No unnatural feature duplication or copy-move cloning detected.
Typography, Font & Baseline Alignment Module	PASS	LOW	94.0%	Typography, font geometry, and baseline alignment are homogeneous across all lines.
Color Space & Illumination Gradient Module	PASS	LOW	95.0%	Color channels and illumination gradient vectors demonstrate natural, single-light-source uniformity.
Signature & Official Stamp Verification Module	PASS	LOW	85.0%	Document evaluated without specialized ink stamp overlays.
Metadata & File Structure Forensics Module	PASS	LOW	94.0%	Clean camera/scanner metadata footprint without third-party graphics editor markers.
Security Features, Barcode & MRZ Forensics Module	PASS	LOW	90.0%	No active security barcode or MRZ zone detected on current document page.
Generative AI & Inpainting Spectral Analysis	TAMPERED	HIGH	99.0%	High-frequency 2D FFT spectral anomaly detected (Ratio: 0.89); Periodic frequency peaks indicate Generative AI / Diffusion inpainting artifacts

FORENSIC VISUAL EVIDENCE INSPECTION

The diagram illustrates the process of Error Level Analysis (ELA) on a document. On the left, the 'Original Document Layer' shows a document with a dark header containing the text 'ORIGINAL DOCUMENT LAYER'. Below the header are several horizontal lines representing text. A small rectangular box highlights a portion of the document, labeled 'APPROVED DOCUMENT # 123456789'. On the right, the 'Error Level Analysis (ELA)' results are shown as a large black rectangle, indicating that the document is a high-quality scan with no visible errors.

Suspicious Regions Map

CHEMICAL & PHYSICAL CLEANLINESS

APPROVED AMOUNT \$ 1,000,000.00



LEGAL FORENSIC NOTICE: This verification certificate was generated by an automated multi-module AI & Image Processing Forensic Analysis Pipeline. Confidence percentages and suspicious region bounding boxes represent machine-learning inferences based on mathematical pixel variance, frequency gradients, and sensor noise statistics.